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Academic year 2024-2025 (Odd Sem)

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

Date	7th Nov 2025	Maximum Marks	(10+50) Marks
Course Code	IS353IA	Duration	(30+90) = 120 Min
Sem	V Sem	CIE-1	
UG/PG	UG	Faculty: AS/VH/VG/SB/SGD/	

Artificial Intelligence and Machine Learning

(Common to CSE/ISE/CD/CY)

(Scheme & Solution)

Q. No.	QUIZ	M

1.	An empirical test was proposed by Alan Turing which was designed to provide a satisfactory operational definition of Intelligence. List the important capabilities the computer needs to possess to pass Turing test and Total Turing Test.
	• The **Turing Test**, proposed by Alan Turing (1950), was designed to provide a satisfactory operational definition of intelligence.
	• The computer would need to possess the following capabilities: ---- 1.5M
	  ○ **natural language processing** to enable it to communicate successfully in English;
	  ○ **knowledge representation** to store what it knows or hears;
	  ○ **automated reasoning** to use the stored information to answer questions and to draw new conclusions;
	  ○ **machine learning** to adapt to new circumstances and to detect and extrapolate patterns.
	• However, the so-called total Turing Test includes a video signal so that the interrogator can test the subject's perceptual abilities." To pass the **total Turing Test**, the computer will need: ---- 0.5M
	  ○ **computer vision** to perceive objects, and
	  ○ **robotics** to manipulate objects and move about
	• These six disciplines compose most of AI.
		2

2.	List the four basic kinds of agent programs that embody the principles underlying almost all intelligent systems. Four basic kinds of agent programs that embody the principles underlying almost all intelligent systems: --- 0.5X4 = 2M • Simple reflex agents; • Model-based reflex agents; • Goal-based agents; and • Utility-based agents.	2
3.	Decision tree induction algorithms must provide a method for expressing an attribute test condition and its corresponding outcomes for different attribute types. Indicate with the aid of pictorial representation (considering real-world examples) for nominal and ordinal attributes. **Methods for Expressing Attribute Test Conditions**	2

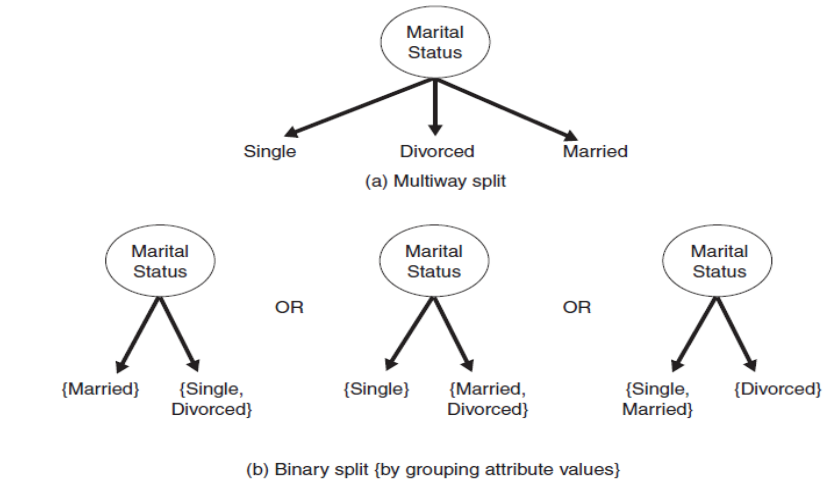


Figure Attribute test conditions for nominal attributes.

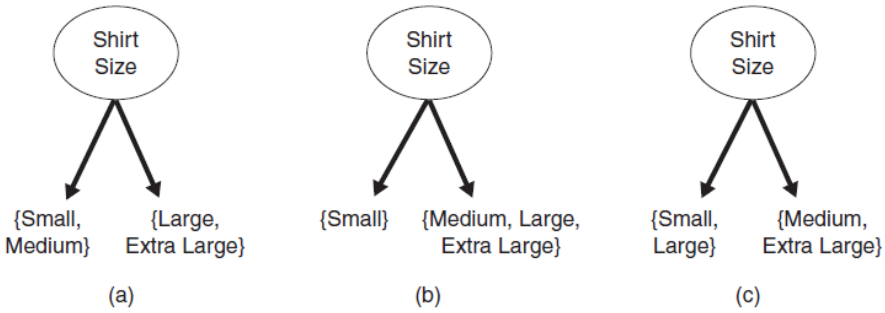
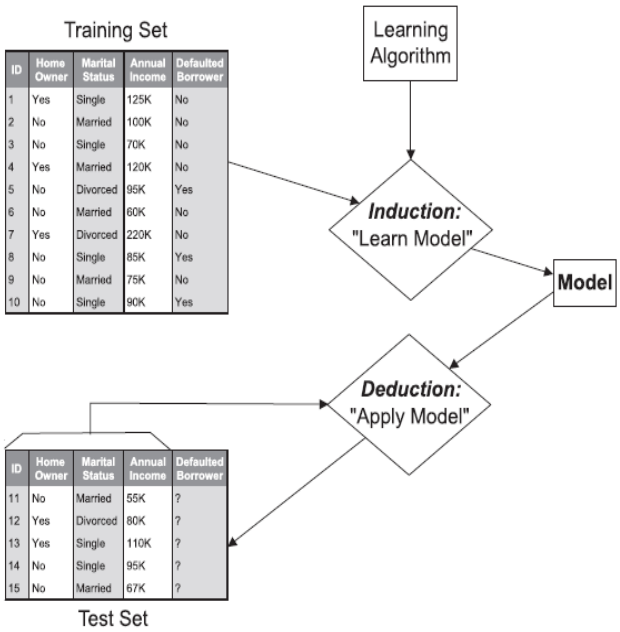


Figure Different ways of grouping ordinal attribute values.

4. Draw the General framework for building a classification model. 2



ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
11	No	Married	55K	?
12	Yes	Divorced	80K	?
13	Yes	Single	110K	?
14	No	Single	95K	?
15	No	Married	67K	?

Figure General framework for building a classification model.

5.	Determine the different impurity measures for evaluating a Single Node given below: Node N_2 | Count Class=0 | 1 Class=1 | 5 $\text{Gini} = 1 - (1/6)^2 - (5/6)^2 = 0.278$ $\text{Entropy} = -(1/6)\log_2(1/6) - (5/6)\log_2(5/6) = 0.650$ $\text{Error} = 1 - \max[1/6, 5/6] = 0.167$	2

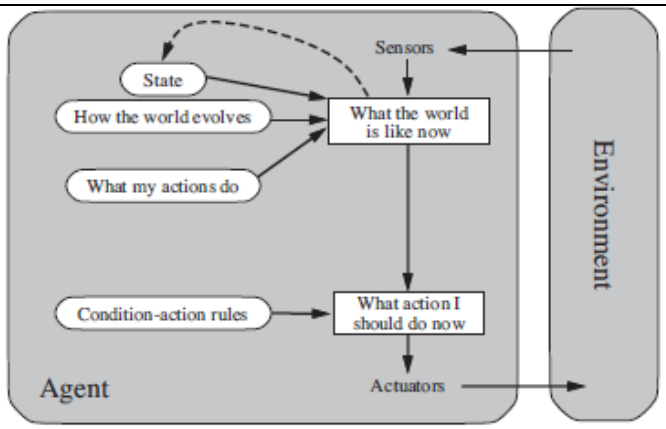
Q. No.	TEST	M
1a.	Summarize in a tabular form the PEAS description of the task environment for Part-picking robot. Identify and list the dimensions (properties) along which task environment of Part-picking robot can be categorized. Justify your answer Summarize in a tabular form the PEAS description of the task environment for Part-picking robot. **----2.5M** Identify and list the dimensions (properties) along which task environment of Part-picking robot can be categorized. Justify your answer. **--- 2.5M** **PEAS description of the task environment for Part-picking robot.**	05

Agent Type	Performance Measure	Environment	Actuators	Sensors
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors

Task environment of Part-picking robot and its characteristics.

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous

1b.	Draw the schematic diagram indicating Structure of Model-based Reflex Agent and discuss the characteristics considering the Scenario of Automated Taxi. Write the Agent program for Model-based Reflex Agent Write the schematic diagram indicating Structure ***of Model-based Reflex Agent*** and discuss the characteristics considering the Scenario of Automated Taxi. ----3M Write the ***Agent program*** for Model-based Reflex Agent. ----- 2M	05



Schematic diagram of a Model-based reflex agent.

```
function MODEL-BASED-REFLEX-AGENT(percept) returns an action
  persistent: state, the agent's current conception of the world state
              model, a description of how the next state depends on current state and action
              rules, a set of condition–action rules
              action, the most recent action, initially none

  state ← UPDATE-STATE(state, action, percept, model)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```

Agent Program - A model-based reflex agent.
It keeps track of the current state of the world, using an internal model. It then chooses an action in the same way as the reflex agent.

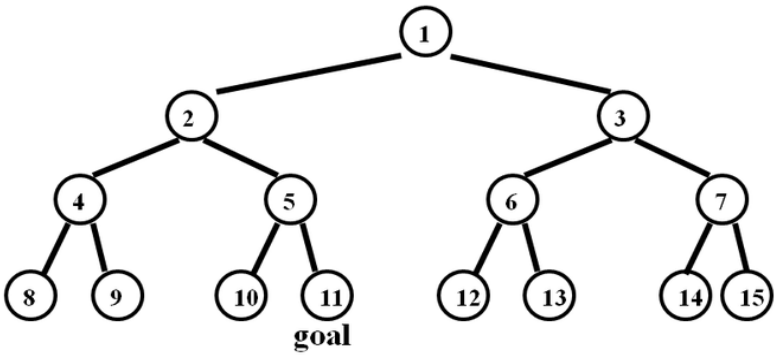
2a. Consider a state space where the start state is number 1 and each state k has two successors: numbers 2k and 2k + 1. 06

i. Draw the portion of the state space for states 1 to 15.

ii. Suppose the goal state is 11. List the order in which nodes will be visited for breadth first search, depth-limited search with limit 3, and iterative deepening search.

Indicate all the intermediate steps representing the solution. **(1.5M +1.5M +2M)**

Answer



Answer

BFS: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

DLS (depth limit 3): 1, 2, 4, 5, 3, 6, 7

IDS:

first iteration: 1

second iteration: 1, 2, 3

third iteration: 1, 2, 4, 5, 3, 6, 7

fourth iteration: 1, 2, 4, 8, 9, 5, 10, 11

2b.	Suppose that an agent lives in a grid world of size 5 x 5 (for a total of 25 squares). The agent has two sensors: a GPS sensor, which informs the agent of its current location on the grid, and a camera sensor, which informs the agent of the color on the current square and the four adjacent squares. The agent, at each step, moves left, right, top, bottom. 24 of the 25 squares are safe, and one square (at location 4,3) is dangerous. The current location of the agent is safe. a. If the agent is reflex-based, the safe squares are green, and the dangerous square is red, is it possible for this agent to follow a safe strategy that will always avoid the dangerous square? If yes, what is that strategy? **----- 2M** **Solution:** Yes, the strategy is to never visit a red square. b. If the agent is reflex-based, and all squares (safe and dangerous) are green, is it possible for this agent to follow a safe strategy that will always avoid the dangerous square? If yes, what is that strategy? **----- 2M** **Solution:** Yes, the strategy is to never visit location (4,3), which can be achieved because the agent has a GPS sensor.	04

3a.	Write the pseudocode for decision tree induction algorithm. Explain the application of decision trees briefly for any real-world use case.
	Write the ***pseudocode for decision tree induction algorithm and explain*** the same briefly. **---- 3M**
	
	Algorithm 3.1 A skeleton decision tree induction algorithm.
	`TreeGrowth` (E, F)
	if `stopping_cond`$(E,F) = true$ then
	  $leaf$ = `createNode`().
	  $leaf.label$ = `Classify`(E).
	  return $leaf$.
	else
	  $root$ = `createNode`().
	  $root.test_cond$ = `find_best_split`(E, F).
	  let $V = \{v \mid v$ is a possible outcome of $root.test_cond\}$.
	  for each $v \in V$ do
	  $E_v = \{e \mid root.test_cond(e) = v$ and $e \in E\}$.
	  $child$ = `TreeGrowth`(E_v, F).
	  add $child$ as descendent of $root$ and label the edge $(root \rightarrow child)$ as v.
	  end for
	end if
	return $root$.
	
	The details of the algorithm:
	(i) The createNode() function extends the decision tree by creating a new node. A node in the decision tree either has a test condition, denoted as *node.test_cond*, or a class label, denoted as *node.label*.
	(ii) The find_best_split() function determines the attribute test condition for partitioning the training instances associated with a node. The splitting attribute chosen depends on the impurity measure used.
	(iii) The Classify() function determines the class label to be assigned to a leaf node. For each leaf node t, let $p(i\|t)$ denote the fraction of training instances from class i associated with the node t. The label assigned to the leaf node is typically the one that occurs most frequently in the training instances that are associated with this node.
	$$leaf.label = \underset{i}{\operatorname{argmax}}\ p(i\|t),$$
	where the argmax operator returns the class i that maximizes $p(i\|t)$.
		05

	(iv) The stopping cond() function is used to terminate the tree-growing process by checking whether all the instances have identical class label or attribute values. **Represent the application of decision tree for any real world problem. ---- 2M**	
3b.	A model is built to detect whether a patient has a disease. The model was tested on 300 patients. Out of 120 patients who actually had the disease, 90 were correctly diagnosed. The model also incorrectly predicted 60 healthy individuals as diseased. Using the above information, compute: i) Confusion matrix ii) Accuracy iii) Error Rate (Assume "Has disease" = Positive class) **Soln:** • Total patients = 300 • Actual positive (Has disease) = 120 • True Positives (TP) = 90 (correctly diagnosed diseased patients) • False Negatives (FN) = 120 - 90 = 30 (missed diseased patients) • False Positives (FP) = 60 (healthy individuals incorrectly predicted as diseased) • Therefore, True Negatives (TN) = 300 - (TP + FN + FP) = 300 - (90 + 30 + 60) = **120** **a) Confusion Matrix -----1M** |	**Actual Diseased (Positive)**	**Actual Healthy (Negative)**
Predicted Diseased	TP = 90	FP = 60
Predicted Healthy	FN = 30	TN = 120 | **b) Accuracy -----2M** Accuracy = (TP + TN) / (TP + TN + FP + FN) = (90 + 120) / 300 = 210 / 300 = **0.70 or 70%** **c) Error Rate -----2M** Error Rate = (FP + FN) / (TP + TN + FP + FN) = (60 + 30) / 300 = 90 / 300 = **0.3 or 30%**	05
4.	Consider the training examples shown in Table 4 for a Binary classification problem. **Table 4** Training Samples for a Binary classification problem	10 (2+4+3+1)

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	−
4	F	F	4.0	+
5	F	T	7.0	−
6	F	T	3.0	−
7	F	F	8.0	−
8	T	F	7.0	+
9	F	T	5.0	−

Determine the following measures that can be used to determine the goodness of an attribute test condition:

(i) What is the entropy of this collection of training examples with respect to the positive class? ---- 2M

There are four positive examples and five negative examples. Thus, $P(+) = 4/9$ and $P(-) = 5/9$. The entropy of the training examples is $-4/9 \log_2(4/9) - 5/9 \log_2(5/9) = 0.9911$.

(ii) What are the information gains of a1 and a2 relative to these training examples? ---- 4M

For attribute a_1, the corresponding counts and probabilities are:

a_1	+	-
T	3	1
F	1	4

The entropy for a_1 is

$$\frac{4}{9}\Big[-(3/4)\log_2(3/4) - (1/4)\log_2(1/4)\Big] + \frac{5}{9}\Big[-(1/5)\log_2(1/5) - (4/5)\log_2(4/5)\Big] = 0.7616.$$

Therefore, the information gain for a_1 is $0.9911 - 0.7616 = 0.2294$.

For attribute a_2, the corresponding counts and probabilities are:

a_2	+	-
T	2	3
F	2	2

The entropy for a_2 is

$$\frac{5}{9}\Big[-(2/5)\log_2(2/5) - (3/5)\log_2(3/5)\Big] + \frac{4}{9}\Big[-(2/4)\log_2(2/4) - (2/4)\log_2(2/4)\Big] = 0.9839.$$

Therefore, the information gain for a_2 is $0.9911 - 0.9839 = 0.0072$.

(iii) For a3, which is a continuous attribute, compute the information gain for every possible split. ---------- 3M

a_3	Class label	Split point	Entropy	Info Gain
1.0	+	2.0	0.8484	0.1427
3.0	-	3.5	0.9885	0.0026
4.0	+	4.5	0.9183	0.0728
5.0	-			
5.0	-	5.5	0.9839	0.0072
6.0	+	6.5	0.9728	0.0183
7.0	+			
7.0	-	7.5	0.8889	0.1022

The best split for a_3 occurs at split point equals to 2.

(iv) What is the best split (among a1, a2, and a3) according to the information gain? ------ 1M

According to information gain, $a1$ produces the best split.

5a. **(i) What is the best split (between $a1$ and $a2$) according to the classification error rate? ---- 2M** — 05 (3+2)

Answer:

For attribute $a1$: error rate = 2/9.

For attribute $a2$: error rate = 4/9.

Therefore, according to error rate, $a1$ produces the best split.

(ii) What is the best split (between $a1$ and $a2$) according to the Gini index? ------ 3M

Answer: For attribute $a1$, the gini index is

$$\frac{4}{9}\left[1-(3/4)^2-(1/4)^2\right]+\frac{5}{9}\left[1-(1/5)^2-(4/5)^2\right]=0.3444.$$

For attribute $a2$, the gini index is

$$\frac{5}{9}\left[1-(2/5)^2-(3/5)^2\right]+\frac{4}{9}\left[1-(2/4)^2-(2/4)^2\right]=0.4889.$$

Since the gini index for $a1$ is smaller, it produces the better split.

5b. Summarize any 5 important characteristics of decision tree induction algorithms. — 05

Important characteristics of decision tree induction algorithms(Any 5) ---- 1 X 5 = 5M

- Applicability
- Expressiveness
- Computational Efficiency
- Handling Missing Values
- Handling Interactions among Attributes
- Handling Irrelevant Attributes
- Handling Redundant Attributes
- Using Rectilinear Splits
- Choice of Impurity Measure

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